i) $K_c = Y\sigma_f\sqrt{\pi a_c}$

$$= \left(\frac{W}{\pi a}\tan\frac{\pi a}{W}\right)^{\frac{1}{2}}\sigma_f\sqrt{\pi a_c}$$

$$= \left(\frac{100\times10^{-3}}{\pi\left(\frac{25}{2}\times10^{-3}\right)}\tan\left(\frac{\pi\left(\frac{25}{2}\times10^{-3}\right)}{100\times10^{-3}}\right)\right)^{\frac{1}{2}}(200\times10^{6})$$

$$\times\sqrt{\pi\left(\frac{25}{2}\times10^{-3}\right)}$$

$$= 1.027027836(200\times10^{6})\sqrt{\pi\left(\frac{25}{2}\times10^{-3}\right)}$$

$$= 40.70447456\times10^{6}\ \text{Pa}\sqrt{\text{m}}$$

$$\approx 40.7\ \text{MPa}\sqrt{\text{m}}$$

ii) For plane stress,

$$r_p = \frac{1}{2\pi}\left(\frac{K}{\sigma_Y}\right)^2$$

$$= \frac{1}{2\pi}\left(\frac{40.7\times10^{6}}{400\times10^{6}}\right)^2$$

$$= 1.648103398\times10^{-3}\ \text{m}$$

$$\approx 1.65\times10^{-3}\ \text{m}$$

$$K_c = Y\sigma\sqrt{\pi(a+r_p)}$$

$$= 1.027027836(200\times10^{6})\sqrt{\pi\left(\frac{25}{2}\times10^{-3}+1.65\times10^{-3}\right)}$$

$$= 43.30482214\times10^{6}\ \text{Pa}\sqrt{\text{m}}$$

$$\approx 43.3\ \text{MPa}\sqrt{\text{m}}$$

6) $\sigma = 200 MPa$ $\sigma_Y = 400 MPa$

For a large sheet, $Y = 1$

i) $K_c = Y\sigma\sqrt{\pi a}$

$$\left(\frac{K_c}{Y\sigma}\right)^2 = \pi a$$

$$a = \frac{K_c^2}{Y^2\sigma^2\pi}$$

$$= \frac{(42\times10^6)^2}{1^2(200\times10^6)^2\pi}$$

$$= 14.03746598\times10^{-3} m$$

$$\approx 14.04 mm$$

$$\therefore 2a = 28.07493196\times10^{-3}$$

$$\approx 28.07 mm$$

ii) $r_p = \frac{1}{2\pi}\left(\frac{K}{\sigma_Y}\right)^2$

$$= \frac{1}{2\pi}\left(\frac{42\times10^6}{400\times10^6}\right)^2$$

$$= 1.754683248\times10^{-3} m$$

$$\approx 1.75 mm$$

$a + r_p = 14.04$

$a = 14.04 - 1.75$

$= 12.28278273 mm$

$\approx 12.28 mm$

$\therefore 2a = 24.56556547 mm$

$\approx 24.57 mm$

3 i) Initial energy $= mgh$

$= 4(9.81)(200\times10^{-3}\sin 40°)$

$= 5.044597161\,J$

$\approx 5.04\,J$

Final energy $= mgh$

$= 4(9.81)(200\times10^{-3}\sin 39°)$

$= 4.938906429\,J$

$\approx 4.94\,J$

Energy losses $= 5.04 - 4.94$

$= 0.105690731\,9$

$\approx 0.11\,J$

ii) Final energy with specimen $= mgh$

$= 4(9.81)(-200\times10^{-3}\sin 1°)$

$= -0.1369664857\,J$

$\approx -0.14\,J$

Energy absorbed in breaking the specimen

$= 4.94 - (-0.14)$

$= 5.075872915\,J$

$\approx 5.08\,J$

4) $S_a = \dfrac{350-(-500)}{2}$

$= 425\,MPa$

$S_r = 350-(-500)$

$= 850\,MPa$

$S_m = \dfrac{350+(-500)}{2}$

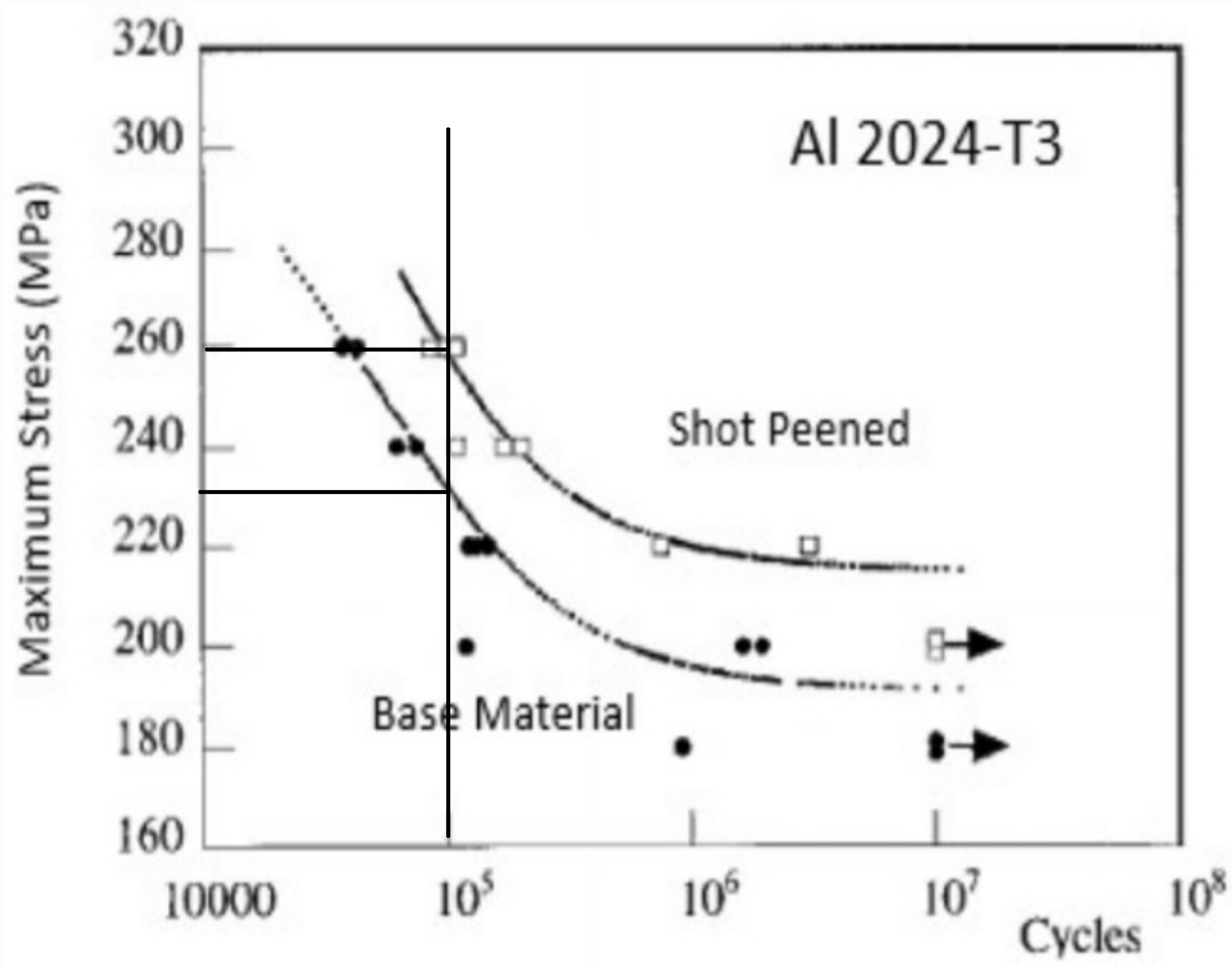
$= -75\,MPa$

$R = \dfrac{\sigma_{min}}{\sigma_{max}} = \dfrac{-500}{350} = -\dfrac{10}{7} \approx -1.429$

5) Number of cycles = 20×5000

$= 10^5$ cycles

From the graph for the base material:



The maximum stress is 230 MPa.

$$\sigma = \frac{F}{A}$$

$$230 \times 10^6 = \frac{600 \times 10^3}{\frac{\pi d^2}{4}}$$

$$d^2 = 3.321494465 \times 10^{-3}$$

$$d = 0.0576324081 \text{ m}$$

$$\approx 57.63 \text{ mm}$$

5) Shot peening is a cold working process used to produce a compressive residual stress layer and modify mechanical properties of metals. It entails impacting a suface with shot (round metallic, glass, or ceramic particles) with force sufficient to create plastic deformation.

Stress for base material = 230MPa

Stress for shot peened material = 260MPa

$$\sigma = \frac{F}{A}$$

$$A_s = \frac{F}{\sigma}$$

$$A_s = \frac{600\times10^3}{260\times10^6}$$

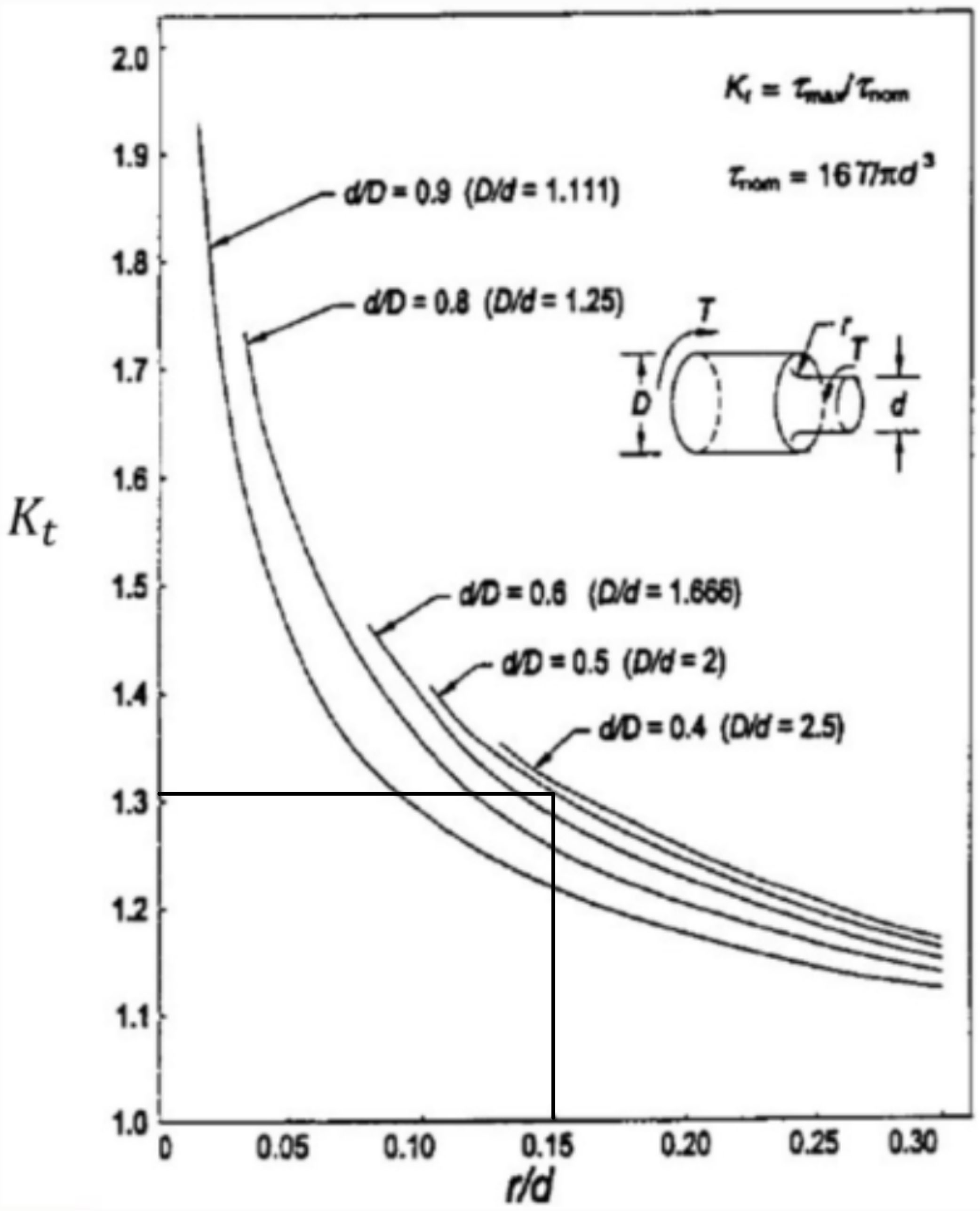
$$= \frac{3}{1300}$$

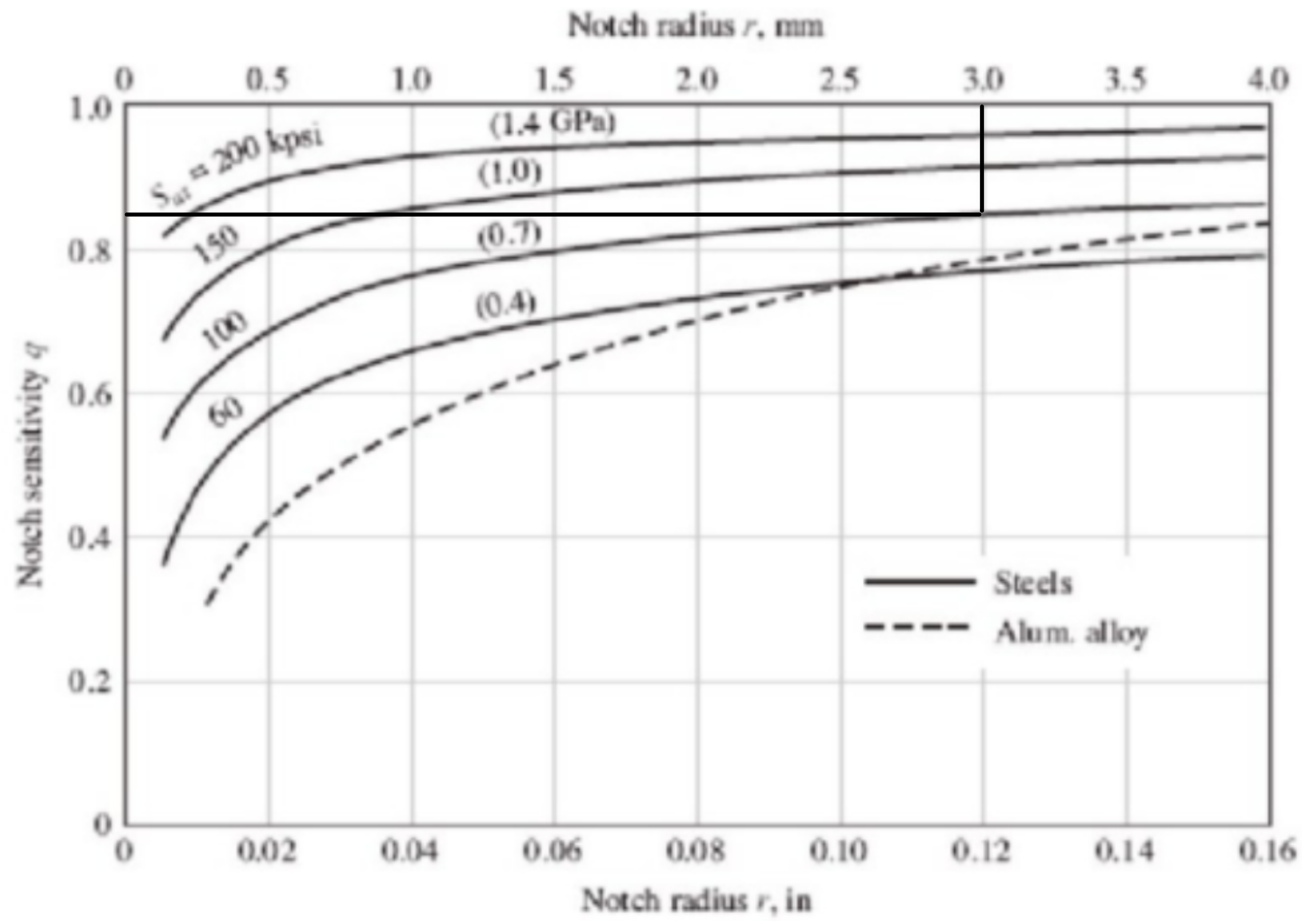
$$A_b = \frac{600\times10^3}{260\times10^6} = \frac{3}{1150}$$

$$\text{Weight saving} = \frac{\frac{3}{1150} - \frac{3}{1300}}{\frac{3}{1150}} = 11.53846154\%$$

$$\approx 11.5\%$$

6)





$T = 30\,Nm$

$d = 20\,mm$

$D = 40\,mm$

$r = 3\,mm$

$\frac{r}{d} = 0.15$

$\frac{d}{D} = 0.5$

From the graphs,

$K_t \approx 1.31$

$q \approx 0.85$

$$\therefore K_f = q(K_t - 1) + 1$$

$$= 0.85(1.31 - 1) + 1$$

$$= 1.2635$$

$$\approx 1.26$$